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Assignment-1 : DBMS

Unit-1

Section A: Introduction to DBMS

1. What is a Database Management System (DBMS)? Why is it required?

Ans: A Database Management system (DBMS) is a software that stores, manages, & retrieves data in an organized way.

It is required to:

- Reduce data redundancy
- Maintain consistency
- Provide security
- Allow multi-user access
- Support backup & recovery.

2. Why is DBMS better than the traditional file system?

Ans. Traditional File system has:

(i) High redundancy
(ii) Data inconsistency
(iii) No proper security
(iv) Difficult data access

While DBMS:

(i) Controls redundancy
(ii) Ensures consistency
(iii) Provides authorization
(iv) Supports concourrency
(v) Has backup & recovery

3. What is data abstraction? Explain its levels.

Ans: Data abstraction means hiding complex internal details from users.

Three levels:

- Physical Level - How data is stored in disk
- Logical Level - What data is stored & relationships
- view Level - What user sees.

Purpose is to simplify database usage.

4. What is the three-schema architecture? Why is it important?

Ans: It consists of:

- Internal schema (Physical Level)
- Conceptual schema (Logical Level)
- External schema (user view level)

Importance:

- Provides data independence
- separates user view from physical storage.

5. What is data independence? Why is it needed?

Ans: Ability to change schema at one level without affecting others.

Types:

- Physical Data Independence - Change storage without affecting logical level.
- Logical Data Independence - Change logical schema without affecting user views.

Needed for flexibility & maintenance.

6. What is an Entity-Relationship (ER) model?

Ans: ER Model is a conceptual model used to design databases.

It represents;

- Entities (real-world objects)
- Attributes (properties)
- Relationships (connections between entities)

Used before database implementation.

CamScanner

4. What are integrity constants? Why they are important in DBMS?

Ans: Rules that ensures data accuracy & reliability.

Types:

- Primary key
- Foreign key
- Unique
- Not NULL
- Check

They are important because they prevent invalid or duplicate data.

Section B: Design, Draw & Defend

1. Draw the 3-schema architecture & explain each level.

Ans:

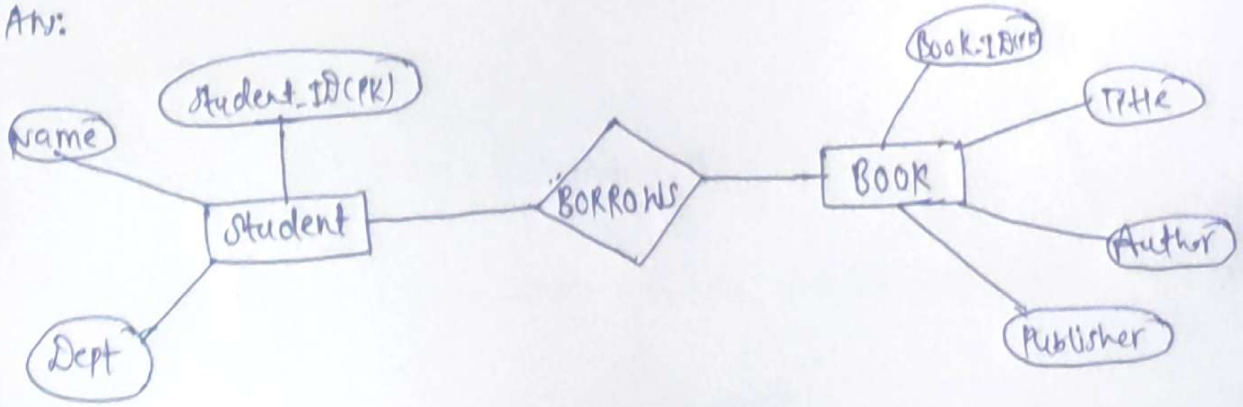
External Level
(User views / Applications)
↓
Conceptual Level
(Logical Database schema)
↓
Internal Level
(Physical storage level)

Explanation:

- External: User-specific views → Ensures data independence.
- Conceptual: overall database design
- Internal: physical storage details

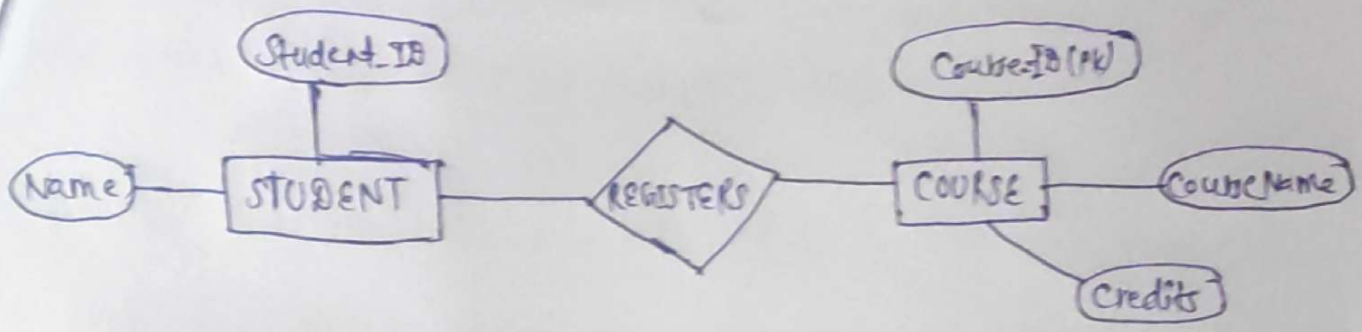
2. Draw an ER diagram for a Library Management System.

Ans:



Draw an ER diagram for a Student-Course Registration System.

Ans:



Section C: Case-Based Analytical Questions:

A. A hospital wants to store patient, doctor, & appointment data. Identify entities, attributes, & relationships.

Ans: Entities:

- Patient (PatientID PK, Name, Age)
- Doctor (DoctorID PK, Specialization)
- Appointment (AppID PK, Date, Time)

Relationships:

- Patient books Appointment
- Doctor attends Appointment.

B. An online shopping platform stores customer, product, & order data. Suggest keys and constraints.

Ans: Entities: →
- Customer (CustomerID PK)
- Product (ProductID PK)
- Order (OrderID PK)

Foreign keys: →
- Order → CustomerID
- Order → ProductID

Constraints: →
- NOT NULL
- UNIQUE
- CHECK (Quantity > 0)

C. A university stores students records. Explain how data abstraction helps different users.

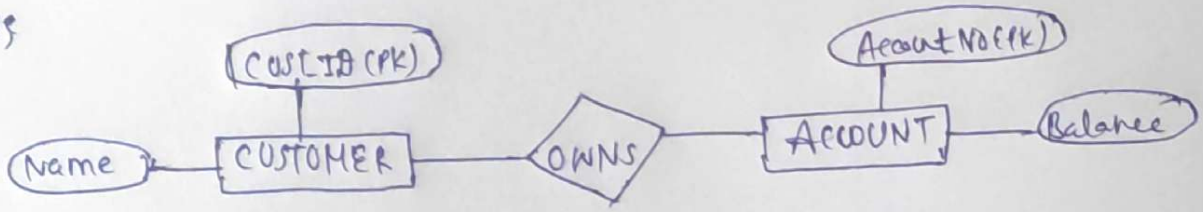
- Student sees marks (View level)
- Admin sees full database structure (Logical level)
- Database admin handles storage (Physical level)

Helps protect sensitive data & simplify usage.

Section D: One Diagram + One Insight (ODONI)

A. Draw a simple ER diagram for any real-life system & explain one key design decision.

Ans:



- Entities & Attributes are given above;
- Relationship : Customer owns Account
- Cardinality : One customer → Multiple Accounts.
- Design Decision : Separate Account entity to allow multiple accounts per customer.

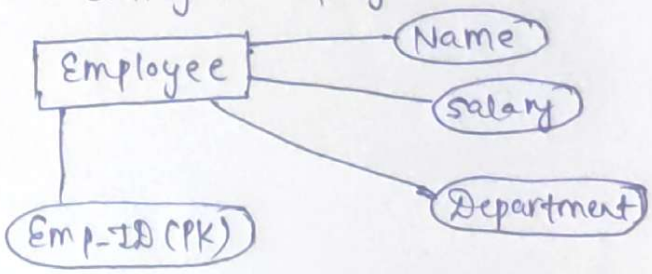
B. Draw the three-schema architecture and mention one benefit.

Ans: External
↓
Conceptual
↓
Internal

Benefits: Changes in storage don't affect user program (Data independence)

C. Draw an entity with attributes & identify its primary key.

Ans: Entity : Employee



Primary key;

→ Emp_ID ensures uniqueness.